Date: 10/09/2025

Lab Practical #15:

Implementation of bit stuffing Using C/Java language with example.

Practical Assignment #15:

1. **C/Java Program: Implementation of Bit stuffing Using C/Java language.**

```
import java.util.Scanner;

public class BitStuffing {

  public static String stuffBits(String data) {
    StringBuilder stuffed = new StringBuilder();
    int count = 0;

    for (char bit : data.toCharArray()) {
      stuffed.append(bit);
      if (bit == '1') {
        count++;
        if (count == 5) {
          stuffed.append('0');
          count = 0;
        }
      } else {
        count = 0;
      }
    }

    return stuffed.toString();
  }
```

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```
public static String destuffBits(String stuffedData) {
    StringBuilder destuffed = new StringBuilder();
    int count = 0;

    for (int i = 0; i < stuffedData.length(); i++) {
        char bit = stuffedData.charAt(i);
        destuffed.append(bit);

        if (bit == '1') {
            count++;
            if (count == 5 && i + 1 < stuffedData.length() && stuffedData.charAt(i + 1) ==
'0') {
                i++;
                count = 0;
            }
        } else {
            count = 0;
        }
    }

    return destuffed.toString();
}

public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);

    System.out.print("Enter binary data: ");
    String data = scanner.nextLine();
```

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```
        String stuffed = stuffBits(data);
        System.out.println("Stuffed Data:   " + stuffed);

        String destuffed = destuffBits(stuffed);
        System.out.println("Destuffed Data: " + destuffed);

        scanner.close();
    }
}
```

1. **Enter the binary data: 011111101111110**
   Bit-stuffed data: 01111101011111010

```
Enter binary data: 011111101111110
Stuffed Data:   01111101011111010
Destuffed Data: 011111101111110
```

2. **Enter the binary data: 111110111111**
   Bit-stuffed data: 1 1 1 1 1 0 0 1 1 1 1 1 0 1

```
Enter binary data: 111110111111
Stuffed Data:   11111001111101
Destuffed Data: 111110111111
```